

$J(\rho_k^\tau) \leq J(\rho_0)$). This gives a uniform bound on $\mathcal{F}(\rho_t^\tau)$ and the Dunford-Pettis Theorem below turns the weak convergence $\rho_t^\tau \rightharpoonup \rho_t$ as measures into a weak convergence in L^1 . Look at the term

$$\int f \cdot \nabla V \rho^\tau := \int_0^T dt \int_{\Omega} f(t, x) \cdot \nabla V(x) \rho_t^\tau(x) dx.$$

The integrand in t is uniformly bounded by $\|f\|_{L^\infty} \|\nabla V\|_{L^\infty}$. Hence, we can compute its limit as $\tau \rightarrow 0$ looking at the limit for each fixed t because it is dominated. And, for fixed t , we have weak convergence in L^1 of ρ_t^τ to ρ_t , which, multiplied times a fixed L^∞ function, i.e. $f \cdot \nabla V$, gives the desired limit of the integral. $\square$ $\square$

Box 8.2. – Memo – Equi-integrability and Dunford-Pettis Theorem

Definition - A sequence of function ρ_n is said to be equi-integrable if for every $\varepsilon > 0$ there exists $\delta > 0$ such that on every set A with $|A| \leq \delta$ we have $\int_A \rho_n(x) dx \leq \varepsilon$ for any n .

We can check that equi-integrability is equivalent to a bound $\int f(\rho_n(x)) dx \leq C < +\infty$ for any superlinear function f . Let us only prove that such a bound implies equi-integrability. Indeed, fix $\varepsilon > 0$ and take M such that for all t with $t > M$ we have $f(t)/t > 2C/\varepsilon$ (this is possible thanks to the superlinearity of f). Then we estimate

$$\int_A \rho_n = \int_{A \cap \{\rho_n \leq M\}} \rho_n + \int_{A \cap \{\rho_n > M\}} \frac{\rho_n}{f(\rho_n)} f(\rho_n) dx \leq M|A| + \frac{\varepsilon}{2C} C.$$

It is now enough to take $\delta = \frac{\varepsilon}{2M}$ and we get $\int_A \rho_n(x) dx < \varepsilon$.

Theorem - Suppose that ρ_n is a sequence of probability densities weakly converging as measures to $\rho \in \mathcal{P}(\Omega)$. Suppose that $\int f(\rho_n)$ is bounded for some superlinear function f . Then ρ is also absolutely continuous and the weak convergence also holds in L^1 (i.e. in duality with L^∞ functions, and not only with continuous ones).

Proof - The absolute continuity of ρ is a consequence of the lower semi-continuity of $\mathcal{F}$ defined as $\mathcal{F}(\rho) = \int_{\Omega} f(\rho(x)) dx$ if $\rho \ll \mathcal{L}^d$ ($+\infty$ otherwise).

Moreover, the bound on $\int f(\rho_n(x)) dx$ implies equi-integrability. Fix a test function $\varphi \in L^\infty$ and use the fact that for every $\varepsilon > 0$ there is a $\delta > 0$ with $|A| \leq \delta \Rightarrow \int_A \rho_n(x) dx \leq \varepsilon$ and that, by Lusin Theorem, for this $\delta > 0$ there exists a continuous function $\tilde{\varphi}$ with $\|\tilde{\varphi}\|_{L^\infty} \leq \|\varphi\|_{L^\infty}$ and $\{\varphi \neq \tilde{\varphi}\} \subset A$, where A is open with $|A| < \delta$. We have

$$\left| \int \varphi(\rho_n - \rho) \right| \leq \left| \int \tilde{\varphi}(\rho_n - \rho) \right| + 2\|\varphi\|_{L^\infty} \int_A (\rho_n + \rho).$$

Since A is open and $\rho_n \rightharpoonup \rho$ we have $\int_A \rho(x) dx \leq \liminf \int_A \rho_n(x) dx \leq \varepsilon$, hence

$$\limsup \left| \int \varphi(x) \rho_n(x) dx - \int \varphi(x) \rho(x) dx \right| \leq 0 + 2\|\varphi\|_{L^\infty} \varepsilon,$$

which implies, ε being arbitrary, $\int \varphi(x) \rho_n(x) dx \rightarrow \int \varphi(x) \rho(x) dx$.

8.4 Discussion

8.4.1 EVI, uniqueness and geodesic convexity

In the general theory of Gradient Flows ([17]) in metric spaces, another characterization, different from the EDE, is proposed in order to cope with uniqueness and stability results. It is based on the following observation: if $F : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, then the inequality

$$F(y) \geq F(x) + p \cdot (y - x) \quad \text{for all } y \in \mathbb{R}^d$$

characterizes (by definition) the vectors $p \in \partial F(x)$ and, if $F \in C^1$, it is only satisfied for $p = \nabla F(x)$. Analogously, if F is λ -convex, the inequality that characterizes the gradient is

$$F(y) \geq F(x) + \frac{\lambda}{2} |x - y|^2 + p \cdot (y - x) \quad \text{for all } y \in \mathbb{R}^d.$$

Hence, we can pick a curve $x(t)$ and a point y and compute

$$\frac{d}{dt} \frac{1}{2} |x(t) - y|^2 = (y - x(t)) \cdot (-x'(t)).$$

Consequently, imposing

$$\frac{d}{dt} \frac{1}{2} |x(t) - y|^2 \leq F(y) - F(x(t)) - \frac{\lambda}{2} |x(t) - y|^2,$$

for all y , will be equivalent to $-x'(t) = \nabla F(x(t))$. This will provide a second characterization (called EVI, *Evolution Variational Inequality*) of gradient flows in a metric environment. Indeed, all the terms appearing in the above inequality have a metric counterpart (only squared distances and derivatives w.r.t. time appear). Even if we often forget the dependence on λ , it should be noted that the condition EVI should actually be written as EVI_λ , since it involves a parameter λ , which is a priori arbitrary. Actually, λ -convexity of F is not necessary to define the EVI_λ property, but it will be necessary in order to guarantee the existence of curves which satisfy such a condition. The notion of λ -convexity will hence be crucial also in metric spaces, where it will be rather “ λ -geodesic-convexity”.

The role of the EVI condition in the uniqueness and stability of gradient flows is quite easy to guess. Take two curves, that we call $x(t)$ and $y(s)$, and compute

$$\frac{d}{dt} \frac{1}{2} d(x(t), y(s))^2 \leq F(y(s)) - F(x(t)) - \frac{\lambda}{2} d(x(t), y(s))^2, \quad (8.13)$$

$$\frac{d}{ds} \frac{1}{2} d(x(t), y(s))^2 \leq F(x(t)) - F(y(s)) - \frac{\lambda}{2} d(x(t), y(s))^2. \quad (8.14)$$

If one wants to estimate $E(t) = \frac{1}{2} d(x(t), y(t))^2$, summing up the two above inequalities, after a chain-rule argument for the composition of the function of two variables³ $(t, s) \mapsto \frac{1}{2} d(x(t), y(s))^2$ and of the curve $t \mapsto (t, t)$, gives

$$\frac{d}{dt} E(t) \leq -2\lambda E(t).$$

³Note that a similar argument, based on doubling the variables, is also often performed for the differentiation of W_2^2 along curves in $\mathbb{W}_2$, which we did differently in Section 5.3.5.

By Gronwall Lemma, this provides uniqueness (when $x(0) = y(0)$) and stability.

The situation concerning these two different notions of gradient flows (EVI and EDE) in abstract metric spaces has been clarified by Savaré (in an unpublished note, but the proof can also be found in [10]), who showed that

- All curves which are gradient flows in the EVI sense also satisfy the EDE condition.
- The EDE condition is not in general enough to guarantee uniqueness of the gradient flow. A simple example: take $X = \mathbb{R}^2$ with the ℓ^∞ distance

$$d((x_1, x_2), (y_1, y_2)) = |x_1 - y_1| \vee |x_2 - y_2|,$$

and take $F(x_1, x_2) = x_1$; we can check that any curve $(x_1(t), x_2(t))$ with $x_1'(t) = -1$ and $|x_2'(t)| \leq 1$ satisfies EDE.

- On the other hand, existence of a gradient flow in the EDE sense is quite easy to get, and provable under very mild assumption.
- The EVI condition is in general too strong in order to get existence (in the example above of the ℓ^∞ norm, no EVI gradient flow would exist), but always guarantees uniqueness and stability (w.r.t. initial data).

Also, the existence of EVI gradient flows is itself very restricting on the function F : indeed, it is proven in [135] that, if F is such that from every starting point x_0 there is an EVI_λ gradient flow, then F is necessarily λ -geodesically-convex.

We need to stress that all these abstract considerations on EDE and EVI are mandatory when dealing with a general metric framework, but can often be skipped when dealing with precise applications. For the sake of the applications to PDE that are at the core of this chapter, the point of view which seems to be more concrete is the following: there is an evolution model translated into a PDE, and we want to study this equation (existence; uniqueness and stability; qualitative, regularity and asymptotical properties; numerical simulations). If this PDE happens to be the gradient flow of a certain functional for the W_2 distance, then much information can be obtained by the gradient flow approach (more specifically, the minimizing movement approach via discretized variational problems). But the notion of solution that one should use is a priori the one which came with the PDE in the modeling of the problem (in general, the notion of weak solution of a continuity equation). This is why, even if (non-trivial) results exist proving equivalence between some abstract (EVI or EDE) notions in the metric space $\mathbb{W}_2$ and weak solutions of evolution equations (see [17]), we preferred not to enter into these details here in this book.

On the other hand, we will see in a while (next section) that, as a last remark in favor of the general theory developed in [17], it is not always easy to prove existence via the Minimizing Movement method when we face non-linear terms and that exploiting the characterization of the gradient flows in $\mathbb{W}_2$ developed in [17] could be sometimes easier.

8.4.2 Other gradient-flow PDEs

We saw in Section 8.3 the example, and a detailed analysis, of the Fokker-Planck equation. The main reason to choose such example is its simplicity, because it is a linear equation. This allows easily to pass to the limit all the terms in the relation between E and ρ . Yet, many other important equations can be obtained as gradient flows in $\mathbb{W}_2$, choosing other functionals. We will see some of them here, without entering into details of the proof. We stress that handling non-linear terms is often difficult and requires ad-hoc estimates. We will discuss some of the ideas that one should apply. Note on the other hand that, if one uses the abstract theory of [17], there is no need to produce these ad-hoc estimates: after developing a general (and hard) theory for general metric spaces, the second part of [17] explains which are the curves that one finds as gradient flows in $\mathbb{W}_2$, with the relation between the velocity field $\mathbf{v}$ and the derivatives (with an ad-hoc notion of subdifferential in the Wasserstein space) of the functional F . This automatically gives the desired result as a part of a larger theory.

We will discuss six classes of PDEs in this section: the *porous media equation*, the *Keller-Segel equation*, more general *aggregation models* and more involved equations with *diffusion, advection and aggregation*, a model for *crowd motion* with density constraints, and the *flow of the squared distance* SW_2^2 .

Porous Media Equation This equation models the diffusion of a substance into a material whose properties are different than the void, and which slows down the diffusion. If one consider the case of particles which are advected by a potential and subject to this kind of diffusion, the PDE reads

$$\partial_t \rho - \Delta(\rho^m) - \nabla \cdot (\rho \nabla V) = 0,$$

for an exponent $m > 1$. One can formally check that this is the equation of the gradient flow of the energy

$$F(\rho) = \frac{1}{m-1} \int \rho^m(x) dx + \int V(x) \rho(x) dx$$

(set to $+\infty$ for $\rho \notin L^m$). Indeed, the first variation of the first part of the functional is $\frac{m}{m-1} \rho^{m-1}$, and $\rho \nabla \left(\frac{m}{m-1} \rho^{m-1} \right) = m \rho \cdot \rho^{m-2} \nabla \rho = \nabla(\rho^m)$.

Note that, in the discrete step $\min_\rho F(\rho) + \frac{W_2^2(\rho, \rho_0)}{2\tau}$, the solution ρ satisfies

$$\begin{cases} \frac{m}{m-1} \rho^{m-1} + V + \frac{\varphi}{\tau} = C & \rho - \text{a.e.} \\ \frac{m}{m-1} \rho^{m-1} + V + \frac{\varphi}{\tau} \geq C & \text{on } \{\rho = 0\}. \end{cases}$$

This allows to express $\rho^{m-1} = \frac{m-1}{m} (C - V - \varphi/\tau)_+$ (see Section 7.2.3). This implies that ρ is compactly supported if ρ_0 is compactly supported, as soon as V has some growth conditions (see Ex(52)). This fact contradicts the usual infinite propagation speed that one finds in linear diffusion models (Heat and Fokker-Planck equation).

The above analysis works in the case $m > 1$: the fact that the usual Fokker-Planck equation can be obtained for $m \rightarrow 1$ can be seen in the following way: nothing changes

if we define F via $F(\rho) = \frac{1}{m-1} \int (\rho^m(x) - \rho(x)) dx + \int V(x)\rho(x) dx$, since the mass $\int \rho(x) dx = 1$ is a given constant. Yet, then it is easy to guess the limit, since

$$\lim_{m \rightarrow 1} \frac{\rho^m - \rho}{m-1} = \rho \log \rho,$$

which provides the entropy that we already used in Section 8.3.

It is also interesting to consider the case $m < 1$: the function $\rho^m - \rho$ is no longer convex, but it is concave and the negative coefficient $1/(m-1)$ makes it a convex function. Unfortunately, it is not superlinear at infinity, which makes it more difficult to handle. But for $m \geq 1 - 1/d$ the functional F is still displacement convex. The PDE that we get as a gradient flow is called *Fast diffusion* equation, and it has different (and opposite) properties in terms of diffusion rate than the porous media one.

From a technical point of view, proving compactness of the minimizing movement scheme for these equations is not very easy, since one needs to pass to the limit the non-linear term $\Delta(\rho^m)$, which means, proving strong convergence on ρ instead of weak convergence. The main ingredient is a sort of H^1 bound in space, which comes from the fact that we have

$$\int_0^T \int_{\Omega} |\nabla(\rho^{m-1})|^2 dx dt \approx \int_0^T \int_{\Omega} \frac{|\nabla \varphi|^2}{\tau^2} dx dt = \int_0^T |\rho'(t)|^2 dt \leq C \quad (8.15)$$

(but one has to deal with the fact that this is not a full H^1 bound, and the behavior in time has to be controlled).

Keller-Segel An interesting model in mathematical biology (see [202, 203] for the original modeling) is the following: a population ρ of bacteria evolves in time, following diffusion and advection by a potential. The potential is given by the concentration u of a chemo-attractant nutrient substance, produced by the bacteria themselves. This kind of phenomenon is also known under the name of *chemotaxis*. More precisely, bacteria move (with diffusion) in the direction where they find more nutrient, i.e. in the direction of ∇u , where the distribution of u depends on their density ρ . The easiest model uses linear diffusion and supposes that the distribution of u is related to ρ by the condition $-\Delta u = \rho$, with Dirichlet boundary conditions $u = 0$ on $\partial\Omega$. This gives the system

$$\begin{cases} \partial_t \rho + \alpha \nabla \cdot (\rho \nabla u) - \Delta \rho = 0, \\ -\Delta u = \rho, \\ u = 0 \text{ on } \partial\Omega, \rho(0, \cdot) = \rho_0, \rho(\partial_n \rho - \partial_n u) = 0 \text{ on } \partial\Omega. \end{cases}$$

The parameter α stands for the attraction intensity of bacteria towards the chemo-attractant. By scaling, instead of using probability measures $\rho \in \mathcal{P}(\Omega)$ one can set $\alpha = 1$ and play on the mass of ρ (indeed, the non-linearity is only in the term $\rho \nabla u$, which is quadratic in ρ).

Alternative equations can be considered for u , such as $-\Delta u + u = \rho$ with Neumann boundary conditions. On the contrary, the boundary conditions on ρ must be of Neumann type, to guarantee conservation of the mass (see next Section 8.4.3). This system can also be set in the whole space, with suitable decay conditions at infinity. Note

also that often the PDE condition defining u as the solution of a Poisson equation is replaced, when $\Omega = \mathbb{R}^2$, by the explicit formula

$$u(x) = -\frac{1}{2\pi} \int_{\mathbb{R}^2} \log(|x-y|) \rho(y) dy. \quad (8.16)$$

There is some confusion in higher dimension, as the very same formula does not hold for the Poisson equation (the logarithmic kernel should indeed be replaced by the corresponding Green function), and there two alternatives: either keep the fact that u solves $-\Delta u = \rho$, or the fact that it derives from ρ through (8.16). We prefer keeping the PDE definition, which sounds more justified.

One can see that this equation is the gradient flow of the functional

$$F(\rho) = \int_{\Omega} \rho \log \rho - \frac{1}{2} \int_{\Omega} |\nabla u_{\rho}|^2, \quad \text{where } u_{\rho} \in H_0^1(\Omega) \text{ solves } -\Delta u_{\rho} = \rho.$$

Indeed, the only non-standard computation is that of the first variation of the Dirichlet term $-\frac{1}{2} \int |\nabla u_{\rho}|^2$. Suppose $\rho_{\varepsilon} = \rho + \varepsilon \chi$ and set $u_{\rho+\varepsilon\chi} = u_{\rho} + \varepsilon u_{\chi}$. Then

$$\frac{d}{d\varepsilon} \left(-\frac{1}{2} \int |\nabla u_{\rho+\varepsilon\chi}|^2 \right)_{\varepsilon=0} = - \int \nabla u_{\rho} \cdot \nabla u_{\chi} = \int u_{\rho} \Delta u_{\chi} = - \int u_{\rho} \chi.$$

It is interesting to note that this Dirichlet term is indeed (up to the coefficient $-1/2$) the square of the H^{-1} norm of ρ , since $\|u\|_{H_0^1} = \|\nabla u\|_{L^2} = \|\rho\|_{H^{-1}}$. We will call it the H^{-1} term.

It is also possible to replace linear diffusion with non-linear diffusion of porous media type, replacing the entropy $\int \rho \log \rho$ with a power-like energy $\int \rho^m$.

Note that the variational problem $\min F(\rho) + \frac{W_2^2(\rho, \rho_0)}{2\tau}$ requires some assumption to admit existence of minimizers, as unfortunately the Dirichlet term has the wrong sign. In particular, it would be possible that the infimum is $-\infty$, or that the energy is not l.s.c. (the H^1 part being u.s.c. because of the negative sign).

When we use non-linear diffusion with $m > 2$ the existence of a solution is quite easy (see Ex(53)). Sophisticated functional inequalities allow to handle smaller exponents, and even the linear diffusion case in dimension 2, provided $\alpha \leq 8\pi$. We refer to [59] and to the references therein for details on the analysis of this equation. Some of the technical difficulties are similar to those of the porous media equation, when passing to the limit non-linear terms. In [59], the H^{-1} term is treated in terms of its logarithmic kernel, and ad-hoc variables symmetrization tricks are used. Note however that the nonlinear diffusion case is easier, as L^m bounds on ρ translate into $W^{2,m}$ bounds on u , and hence strong compactness for ∇u .

We also remark that the above model, coupling a parabolic equation on ρ and an elliptic one on u , implicitly assumes that the configuration of the chemoattractant instantaneously follows that of ρ . More sophisticated models can be expressed in terms of the so-called *parabolic-parabolic Keller-Segel* equation, in the form

$$\begin{cases} \partial_t \rho + \alpha \nabla \cdot (\rho \nabla u) - \Delta \rho = 0, \\ \partial_t u - \Delta u = \rho, \\ u = 0 \text{ on } \partial\Omega, \rho(0, \cdot) = \rho_0, \rho(\partial_n \rho - \partial_n u) = 0 \text{ on } \partial\Omega. \end{cases}$$

or other variants with different boundary conditions. This equation can also be studied as a gradient flow in two variables, using distance W_2 on ρ and L^2 on u ; see [60].

Aggregation models Consider a more general case where the movement is advected by a potential determined by the superposition of many potentials, each created by one particle. For instance, given a function $W : \mathbb{R}^d \rightarrow \mathbb{R}$, the particle located at x produces a potential $W(\cdot - x)$ and, globally, the potential is given by $V(y) = \int W(y - x) d\rho(x)$, i.e. $V = W * \rho$. The equation, if every particles follows $-\nabla V$ is

$$\partial_t \rho - \nabla \cdot (\rho ((\nabla W) * \rho)) = 0,$$

where we used $\nabla(W * \rho) = (\nabla W) * \rho$. If W is even (i.e. the interaction between x and y is the same as between y and x), then this is the gradient flow of the functional

$$F(\rho) = \frac{1}{2} \iint W(x - y) d\rho(x) d\rho(y).$$

When W is convex, for instance in the quadratic case $\iint |x - y|^2 d\rho(x) d\rho(y)$, this gives raise to a general aggregation behavior of the particle, and as $t \rightarrow \infty$ one expects $\rho_t \rightarrow \delta_{x_0}$ (the point x_0 depending on the initial datum ρ_0 : in the quadratic example above it is the barycenter of ρ_0 , see **Ex(51)**). If W is not smooth enough, the aggregation into a unique point can also occur in finite time, see [117].

Obviously, more general interaction terms can be considered, of the form $W(x, y)$. As we saw in Chapter 7, some other energies (as the H^{-1} term in Keller-Segel) can also be expressed in this way, using their Green function (but in general they lose convexity of the kernel). Note that these equations are both non-linear (the term in the divergence is quadratic in ρ) and non-local. It is rare to see these non-local aggregation terms alone in the equation, as they are often coupled with diffusion or other terms. This is why we do not provide specific references except [117]. We also note that from the technical point of view this kind of nonlinearity is much more compact than the previous ones, since the convolution operator transforms weak convergence into strong one, provided W is regular enough (the difficulties with the kernel in Keller-Segel exactly come from its singularity).

Diffusion, advection and aggregation: the full picture The aggregation energy of the previous paragraph are often studied together with an internal energy and a confining potential energy, using the functional

$$F(\rho) = \int f(\rho(x)) dx + \int V(x) d\rho(x) + \frac{1}{2} \iint W(x - y) d\rho(x) d\rho(y).$$

This gives the equation

$$\partial_t \rho - \nabla \cdot (\rho [\nabla(f'(\rho)) + \nabla V + (\nabla W) * \rho]) = 0.$$

Among the mathematical interest for this family of equations, we stress that they are those where more results (in termes of stability, and convergence to equilibrium) can be proven, due to the fact that conditions to guarantee that F is displacement convex are well-known (Section 7.3). See in particular [118, 119] for physical considerations and convergence results on this equation.

Crowd motion The theory of Wasserstein gradient flows has interestingly been applied to the study of a continuous model of crowd movement under density constraints.

Let us explain the modeling, starting from the discrete case. Suppose that we have a population of particles such that each of them, if alone, would follow its own velocity u (which could a priori depend on time, position, on the particle itself. . .). Yet, these particles are modeled by rigid disks that cannot overlap, hence, it is not the actual velocity cannot always be u , in particular if u tends to concentrate the masses. We will call $\mathbf{v}$ the actual velocity of each particle, and the main assumption of the model is that $\mathbf{v} = P_{\text{adm}(q)}(u)$, where q is the particle configuration, $\text{adm}(q)$ is the set of velocities that do not induce (for an infinitesimal time) overlapping starting from the configuration q , and $P_{\text{adm}(q)}$ is the projection on this set.

The simplest example is the one where every particle is a disk with the same radius R and center located at q_i . In this case we define the admissible set of configurations K through

$$K := \{q = (q_i)_i \in \Omega^N : |q_i - q_j| \geq 2R \text{ for all } i \neq j\}.$$

In this way the set of admissible velocities is easily seen to be

$$\text{adm}(q) = \{\mathbf{v} = (\mathbf{v}_i)_i : (\mathbf{v}_i - \mathbf{v}_j) \cdot (q_i - q_j) \geq 0 \text{ for all } (i, j) \text{ with } |q_i - q_j| = 2R\}.$$

The evolution equation which has to be solved to follow the motion of q is then

$$q'(t) = P_{\text{adm}(q(t))}u(t) \quad (8.17)$$

(with $q(0)$ given). Equation (8.17), not easy from a mathematical point of view, was studied by Maury and Venel in [227, 225].

We are now interested in the simplest continuous counterpart of this microscopic model (without pretending that it is any kind of homogenized limit of the discrete case, but only an easy re-formulation in a density setting). In this case the particles population will be described by a probability density $\rho \in \mathcal{P}(\Omega)$, the constraint becomes a density constraint $\rho \leq 1$ (we define the set $K = \{\rho \in \mathcal{P}(\Omega) : \rho \leq 1\}$), the set of admissible velocities will be described by the sign of the divergence on the saturated region $\{\rho = 1\}$: $\text{adm}(\rho) = \{\mathbf{v} : \Omega \rightarrow \mathbb{R}^d : \nabla \cdot \mathbf{v} \geq 0 \text{ on } \{\rho = 1\}\}$; we will consider a projection P , which will be either the projection in $L^2(\mathcal{L}^d)$ or in $L^2(\rho)$ (this will turn out to be the same, since the only relevant zone is $\{\rho = 1\}$). Finally, we solve the equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t (P_{\text{adm}(\rho_t)} u_t)) = 0. \quad (8.18)$$

The main difficulty is the fact that the vector field $\mathbf{v} = P_{\text{adm}(\rho_t)} u_t$ is neither regular (since it is obtained as an L^2 projection, and may only be expected to be L^2 a priori), nor it depends regularly on ρ (it is very sensitive to small changes in the values of ρ : for instance passing from a density 1 to a density $1 - \varepsilon$ completely modifies the saturated zone, and hence the admissible set of velocities and the projection onto it).

In [224] these difficulties have been overpassed in the case $u = -\nabla D$ (where $D : \Omega \rightarrow \mathbb{R}$ is a given Lipschitz function) and the existence of a solution (with numerical simulations) is proven via a gradient flow method. Indeed, (8.18) turns out to be the

gradient flow in $\mathbb{W}_2$ of the energy

$$F(\rho) = \begin{cases} \int D d\rho & \text{if } \rho \in K; \\ +\infty & \text{if } \rho \notin K. \end{cases}$$

We do not enter into the details of the study of this equation, but we just make a little bit more precise the definitions above. Actually, instead of considering the divergence of vector fields which are only supposed to be L^2 , it is more convenient to give a better description of $\text{adm}(\rho)$ by duality :

$$\text{adm}(\rho) = \left\{ \mathbf{v} \in L^2(\rho) : \int \mathbf{v} \cdot \nabla p \leq 0 \quad \forall p \in H^1(\Omega) : p \geq 0, p(1-\rho) = 0 \right\}.$$

In this way we characterize $\mathbf{v} = P_{\text{adm}(\rho)}(u)$ through

$$\begin{aligned} u &= \mathbf{v} + \nabla p, \quad \mathbf{v} \in \text{adm}(\rho), \quad \int \mathbf{v} \cdot \nabla p = 0, \\ p &\in \text{press}(\rho) := \{p \in H^1(\Omega), p \geq 0, p(1-\rho) = 0\}, \end{aligned}$$

where $\text{press}(\rho)$ is the space of functions p used as test functions in the dual definition of $\text{adm}(\rho)$, which play the role of a pressure affecting the movement. The two cones $\nabla \text{press}(\rho)$ (defined as the set of gradients of elements of $\text{press}(\rho)$) and $\text{adm}(\rho)$ are in duality for the L^2 scalar product (i.e. one is defined as the set of vectors which make a negative scalar product with all the elements of the other). This allows for an orthogonal decomposition $u_t = \mathbf{v}_t + \nabla p_t$, and gives the alternative expression of Equation (8.18), i.e.

$$\begin{cases} \partial_t \rho_t + \nabla \cdot (\rho_t (u_t - \nabla p_t)) = 0, \\ 0 \leq \rho \leq 1, p \geq 0, p(1-\rho) = 0. \end{cases} \quad (8.19)$$

More details can be found in [224, 265, 226]. In particular, in [224] it is explained how to handle the nonlinearities when passing to the limit. Two sources of nonlinearity are observed: the term $\rho \nabla p$ is easy to consider, since it is actually equal to ∇p (from $p = 0$ on $\{\rho \neq 1\}$); on the other hand, we need to deal with the equality $p(1-\rho) = 0$ and pass it to the limit. This is done by obtaining strong compactness on p , from a bound on $\int_0^T \int_\Omega |\nabla p|^2$, obtained similarly to (8.15).

Sliced Wasserstein distance We already mentioned in Section 2.5.2 the idea by M. Bernot which consisted in moving every particle following the vector field $\mathbf{v}$ obtained in the following way: given two measures $\rho, \nu \in \mathcal{P}(\mathbb{R}^d)$, we project them onto any one-dimensional direction $e \in \mathbb{S}^{d-1}$ via the map $\pi_e : \mathbb{R}^d \rightarrow \mathbb{R}$ given by $\pi_e(x) = x \cdot e$; call $T_e : \mathbb{R} \rightarrow \mathbb{R}$ the monotone optimal transport between the two image measures $(\pi_e)_\# \rho$ and $(\pi_e)_\# \nu$. Then we define $\mathbf{v}_e(x) := (T_e(\pi_e(x)) - \pi_e(x))e$ and $\mathbf{v}(x) = \int_{\mathbb{S}^{d-1}} \mathbf{v}_e(x) d\mathcal{H}^{d-1}(e)$, where $\mathcal{H}^{d-1}$ is the uniform measure on the sphere.

For numerical approaches, Bernot proposed to iterate a construction with a time step $\tau > 0$, but a natural continuous counterpart exists: simply define, for every absolutely continuous measure $\rho \in \mathcal{P}(\mathbb{R}^d)$, the vector field $\mathbf{v}_{(\rho)}$ defined above (absolute

continuity is just required to avoid atoms in the projections). Then, we solve the equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_{(\rho_t)}) = 0.$$

It happens that this equation has a gradient flow structure: it is indeed the gradient flow in $\mathbb{W}_2$ (i.e. for the distance W_2) of the functional

$$F(\rho) := SW_2^2(\rho, \nu).$$

Existence and estimates on the solution of this equation are proven in [66], and the nonlinearity of $\mathbf{v}_{(\rho)}$ is quite easy to deal with. On the other hand, many, natural and useful, questions are still open: is it true that $\rho_t \rightarrow \nu$ as $t \rightarrow \infty$? Can we define (at least under regularity assumptions on the initial data) the flow of the vector field $\mathbf{v}_{(\rho_t)}$, and what is the limit of this flow as $t \rightarrow \infty$? The idea is that it should be a transport map between ρ_0 and ν and, if not the optimal transport map, at least a “good” one. This has been tested in the discrete case (where both ρ_0 and ν are a finite number of equal Dirac masses) with satisfactory numerical results, which we presented briefly in Section 2.5.2.

8.4.3 Dirichlet boundary conditions

For sure, the attentive reader has already noted that all the equations that have been identified as gradient flows for the distance W_2 on a bounded domain Ω are always accompanied by Neumann boundary conditions. This should not be surprising. Wasserstein distances express the movement of masses when passing from a configuration to another, and the equation represents the conservation of mass. It means that we are describing the movement of a collection ρ of particles, bound to stay inside a given domain Ω , and selecting their individual velocity $\mathbf{v}$ in way which is linked to the global value of a certain functional $F(\rho)$. It is natural in this case to have boundary conditions which write down the fact that particles do not exit the domain. The pointwise value of the density ρ at the points of the boundary $\partial\Omega$ is not particularly relevant in this analysis. Note that “do not exit” does not mean “those on the boundary stay on the boundary”, which is what happens when solutions are smooth and the velocity field $\mathbf{v}$ satisfies $\mathbf{v} \cdot \mathbf{n} = 0$. Yet, the correct Neumann condition here is rather $\rho \mathbf{v} \cdot \mathbf{n} = 0$ a.e., which means that particles could enter from $\partial\Omega$ into the interior $\overset{\circ}{\Omega}$, but immediately after it happens there will be (locally) no mass on the boundary, and the condition is not violated, hence. On the contrary, should some mass go from $\overset{\circ}{\Omega}$ to outside Ω , then we would have a violation of the Neumann condition, since there would be (intuitively) some mass $\rho > 0$ on the boundary with velocity directed outwards.

Anyway, we see that Dirichlet conditions do not find their translation into W_2 gradient flows!

To cope with Dirichlet boundary conditions, Figalli and Gigli defined in [166] a sort of modified Wasserstein distance, with a special role played by the boundary $\partial\Omega$, in order to study the Heat equation $\partial_t \rho = \Delta \rho$ with Dirichlet b.c. $\rho = 1$ on $\partial\Omega$.

Their definition is as follows: given two finite positive measures $\mu, \nu \in \mathcal{M}_+(\overset{\circ}{\Omega})$ (not necessarily probabilities, not necessarily with the same mass), we define

$$\Pi b(\mu, \nu) = \{ \gamma \in \mathcal{M}_+(\overline{\Omega} \times \overline{\Omega}) : (\pi_x)_\# \gamma \llcorner \overset{\circ}{\Omega} = \mu, (\pi_y)_\# \gamma \llcorner \overset{\circ}{\Omega} = \nu \}.$$

Then, we define

$$Wb_2(\mu, \nu) := \sqrt{\inf \left\{ \int_{\bar{\Omega} \times \bar{\Omega}} |x - y|^2 d\gamma, \gamma \in \Pi b(\mu, \nu) \right\}}.$$

The index b stands for the special role played by the boundary. Informally, this means that the transport from μ to ν may be done usually (with a part of γ concentrated on $\mathring{\Omega} \times \mathring{\Omega}$), or by moving some mass from μ to $\partial\Omega$ (using $\gamma_{\perp}(\mathring{\Omega} \times \partial\Omega)$), then moving from one point of the boundary to another point of the boundary (this should be done by using $\gamma_{\perp}(\partial\Omega \times \partial\Omega)$, but since this part of γ does not appear in the constraints, then we can forget about it, and the transport is finally free on $\partial\Omega$), and finally from $\partial\Omega$ to ν (using $\gamma_{\perp}(\partial\Omega \times \mathring{\Omega})$).

In [166] the authors prove that Wb_2 is a distance, that the space $\mathcal{M}_+(\mathring{\Omega})$ is always a geodesic space, independently of convexity or connectedness properties of Ω (differently from what happens with Ω , since here the transport is allowed to “teleport” from one part of the boundary to another, either to pass from one connected component to another or to follow a shorter path going out of Ω), and they study the gradient flow, for this distance, of the functional $F(\rho) = \int (\rho \log \rho - \rho) dx$. Note that in the usual study of the entropy on $\mathcal{P}(\Omega)$ one can decide to forget the term $-\int \rho$, which is anyway a constant because the total mass is fixed. Here this term becomes important (if the function $f(t) = t \log t - t$ is usually preferred to $t \log t$, it is because its derivative is simpler, $f'(t) = \log t$, without changing its main properties).

With this choice of the functional and of the distance, the gradient flow that Figalli and Gigli obtain is the Heat equation with the particular boundary condition $\rho = 1$ on $\partial\Omega$. One could wonder where the constant 1 comes from and a reasonable explanation is the following: if the transport on the boundary is free of charge, then automatically the solution selects the value which is the most performant for the functional, i.e. the constant t which minimizes $f(t) = t \log t - t$. In this way, changing the linear part and using $F(\rho) = \int (\rho \log \rho - c\rho) dx$ could change the constant on the boundary, but the constant 0 is forbidden for the moment. It would be interesting to see how far one could go with this approach and which Dirichlet conditions and which equations could be studied in this way, but this does not seem to be done at the moment.

Moreover, the authors explain that, due to the lack of geodesic convexity of the entropy w.r.t. Wb_2 , the standard abstract theory of gradient flows is not able to provide uniqueness results (the lack of convexity is due in some sense to the possible concentration of mass on the boundary, in a way similar to what happened in [224] when dealing with the door on $\partial\Omega$). On the other hand, standard hilbertian results on the Heat equation can provide uniqueness for this equation, as the authors smartly remark in [166].

We observe that this kind of distances with free transport on the boundary were already present in [72, 71], but in the case of the Wasserstein distance W_1 , and the analysis in those papers was not made for applications to gradient flows, which are less

natural to study with $p = 1$. We can also point out a nice duality formula:

$$\begin{aligned} Wb_1(\mu, \nu) &:= \min \left\{ \int |x - y| d\gamma : \gamma \in \Pi b(\mu, \nu) \right\} \\ &= \sup \left\{ \int u d(\mu - \nu) : u \in \text{Lip}_1(\Omega), u = 0 \text{ on } \partial\Omega \right\}. \end{aligned}$$

Note that in general we also have the (easy) equality

$$Wb_p(\mu, \nu) = \inf \{ W_p(\mu + \chi^+, \nu + \chi^-) : \text{spt}(\chi^\pm) \subset \partial\Omega \},$$

where the Wasserstein distance on the right is defined, by abuse of notation, for every pair of positive measures with the same mass, not only for probabilities. In the special case $p = 1$ and $\mu(\mathring{\Omega}) = \nu(\mathring{\Omega})$, we also obtain

$$Wb_1(\mu, \nu) = \mathcal{T}_c(\mu, \nu), \quad \text{for } c(x, y) = \min\{|x - y|, d(x, \partial\Omega) + d(y, \partial\Omega)\}.$$

The cost c is a pseudo-distance on Ω where moving on the boundary is free. This kind of distance has also been used in [98] (inspired by [71]) to model free transport costs on other lower dimensional sets, and not only the boundary (with the goal to model, for instance, transportation networks, and optimize their shape). It is interesting to see the same kind of ideas appear for so different goals.

8.4.4 Evolution PDEs: not only gradient flows

We finish this chapter (and this book) with some other applications and connections of optimal transport with evolution PDEs (note that we already saw connections with static PDEs: the Monge-Ampère equation, in Section 1.7.6, and the Monge-Kantorovich system, in Section 4.2.2).

Incompressible Euler Equation Let us consider an incompressible and homogeneous fluid moving inside a smooth domain Ω . The Euler equation for incompressible fluids describes the evolution of its velocity field $\mathbf{v}$ in terms of the pressure field p :

$$\partial_t \mathbf{v}_t + (\mathbf{v}_t \cdot \nabla) \mathbf{v}_t + \nabla p_t = 0, \quad \nabla \cdot \mathbf{v}_t = 0.$$

To clarify the notation $(\mathbf{v} \cdot \nabla) \mathbf{v}$ we could write componentwise as $\partial_t \mathbf{v}^i + \mathbf{v}^j \mathbf{v}_j^i = p_i$ (we ignore here the subscripts t for time, and use superscripts for components and subscripts for derivatives, and Einstein's convention for summation over repeated indices). Note that the term $\partial_t \mathbf{v}_t + (\mathbf{v}_t \cdot \nabla) \mathbf{v}_t$ is nothing but the acceleration in Lagrangian coordinates. Indeed, the trajectory of a particle initially at position x is obtained by following the vector field $\mathbf{v}_t$, i.e. we have, as usual, the flow $y'_x(t) = \mathbf{v}_t(y_x(t))$ with $y_x(0) = x$. If we denote by Y_t the map $x \mapsto y_x(t)$, for each time t the map Y_t is a measure-preserving diffeomorphism of Ω , and for each x we have

$$y''_x(t) = -\nabla p_t(y_x(t)).$$

In [23], Arnold interpreted the equation above as a geodesic equation on the space $\text{SDiff}(\Omega)$ of measure-preserving diffeomorphism of Ω , viewed as an infinite-dimensional manifold with the metric inherited from the embedding in $L^2(\Omega)$ (the tangent space corresponding to the divergence-free vector fields). According to this interpretation, one can find solutions by minimizing

$$\int_0^T \int_D \frac{1}{2} |\dot{g}(t, x)|^2 dx dt \quad (8.20)$$

among all paths $g(t, \cdot) : [0, T] \rightarrow \text{SDiff}(\Omega)$ with $g(0, \cdot) = g_0$ and $g(T, \cdot) = g_T$ prescribed (typically, by right invariance, g_0 is taken as the identity map). In this way, the pressure field arises as a Lagrange multiplier from the incompressibility constraint.

Shnirelman proved in [284, 285] that when $d \geq 3$ the infimum is not attained in general, and that when $d = 2$ there exists $h \in \text{SDiff}(D)$ which cannot be connected to the identity by a path with finite action. These “negative” results motivate the study of relaxed versions of Arnold’s problem.

The first relaxed version of Arnold’s minimization problem was introduced by Brenier in [84] and it is a sort of transport problem. Its relation to Arnold’s problem is the same that Kantorovich problem has w.r.t. Monge. Brenier considered probability measures Q on the space $\text{AC}(\Omega)$ of absolutely continuous paths (here below simply denoted $\mathcal{C}$) $\omega : [0, T] \rightarrow \Omega$, and looked at the variational problem

Problem 8.11. Solve

$$(\text{EP}) \quad \min \left\{ \int_{\mathcal{C}} K_2(\omega) dQ(\omega) : Q \in \mathcal{P}(\mathcal{C}), (e_0, e_T)_\# Q = \gamma, (e_t)_\# Q = \mathcal{L}_\Omega \forall t \right\}. \quad (8.21)$$

where $K_2(\omega) := \int_0^T \frac{1}{2} |\omega'(t)|^2 dt$ is the kinetic energy of a curve ω and the transport plan $\gamma \in \mathcal{P}(\mathcal{L}_\Omega, \mathcal{L}_\Omega)$ is fixed, typically of the form $\gamma = (\text{id}, h)_\# \mathcal{L}_\Omega$.

The existence of a minimizer Q is standard, provided that there exists at least a Q with finite action (see [84]), which is true when Ω is smooth enough.

We observe that any sufficiently regular path $g(t, \cdot) : [0, 1] \rightarrow \text{SDiff}(\Omega)$ induces a flow (a traffic plan, in our language, see Section 4.2) Q , but the converse is far from being true: particles starting from different points are allowed to cross at a later time, and particles starting from the same point are allowed to split, which is of course forbidden by classical flows. These solutions are called non-deterministic, and [84] proposes indeed an example in the easy case $T = \pi$, $\Omega = B(0, 1)$ and $h(x) = -x$ where they can be optimal.

On the other hand, in [84], a consistency result was proved: smooth solutions to the Euler equation are optimal even in the larger class of admissible Q provided the pressure field p satisfies

$$\sup_{t \in [0, T]} \sup_{x \in \Omega} \nabla_x^2 p(t, x) \leq \frac{\pi^2}{T^2} I \quad (8.22)$$

and are the unique ones if the above inequality is strict.

Uniqueness of the solution of the minimal action problem is an interesting matter: in [53] the situation is analyzed in 1D (when a general uniqueness result is proven

assuming the pressure to be smooth) and 2D (where, on the contrary, [53] extends and characterizes the non-deterministic solutions of [84] finding a very rich family of non-deterministic optimizers).

Optimal transport appears in this framework in two aspects. First, the action-minimization problem by Brenier is indeed a multi-marginal Kantorovich problem, with marginal constraints for each time t (see Section 1.7.4). Second, when we discretize in time this problem, natural strategies for numerics involve the following procedure: suppose we have determined the positions of the particles at time t_{i-1} and t_{i+1} (with a uniform partition $t_i = T \frac{i}{n}$) and that they are represented by two functions X_{i-1} and X_{i+1} , both with image measure equal to $\mathcal{L}_\Omega$; without the incompressibility constraint, the intermediate value $X_i = (X_{i-1} + X_{i+1})/2$ would be optimal in order to minimize the kinetic energy, but its image measure is not, in general, $\mathcal{L}_\Omega$. Hence, we need to project it onto the set of maps X with prescribed image measure, which is a classical transport problem, in the polar factorization formulation (see Section 1.7.2).

Coupling HJ and continuity equations, Mean Field Games We saw in Remark 6.2 in Chapter 6 that the Benamou-Brenier method provides a solution of the following PDE system

$$\begin{cases} \partial_t \rho + \nabla \cdot (\rho \nabla \varphi) = 0 \\ \partial_t \varphi + \frac{1}{2} |\nabla \varphi|^2 = 0, \rho_0 = \mu, \rho_1 = \nu, \end{cases}$$

where the second equation is satisfied ρ -a.e., with inequality $\partial_t \varphi + \frac{1}{2} |\nabla \varphi|^2 \leq 0$ everywhere.

This kind of coupled systems with Hamilton-Jacobi and continuity equations also arise in other situations, and in particular in the newborn theory of Mean Field Games (MFG).

Let us give a quick and informal presentation of this theory. We will only consider the deterministic case (i.e. there will be no stochastic part in the evolution of the agents and we will use deterministic optimal control theory and not stochastic control). The reader may refer to [206, 207] and [106] to have a wider introduction to this theory.

Suppose that a population of agents may evolve in time, each agent following trajectories of the controlled equation

$$y'(t) = f(t, y(t), \alpha(t)), \quad t \in [0, T] \quad (8.23)$$

$\alpha : [0, T] \rightarrow \mathbb{R}^d$ being a control that every agent may choose. At each time t , the goal of each agent is to maximize the payoff

$$-\int_t^T \left(\frac{|\alpha(s)|^2}{2} + g(\rho(s, y(s))) \right) ds + \Phi(y(T)), \quad (8.24)$$

where g is a given increasing function. This means that α is the effort that every agent makes to move in the desired direction, and he pays for it (for simplicity, we take a quadratic cost), that its position depends on α through the equation (8.23), and that he tries to optimize the final payoff Φ but he also pays for the densities of the regions he passes by. In particular agents would like to avoid overcrowded areas. At this first

step the density $\rho(t, x)$ is supposed to be a given function. Yet, the MFG problem is an equilibrium problem: given the initial density ρ_0 , find a time-dependent family of densities ρ_t such that, when every agent selects its optimal trajectory so as to optimize (8.24), the density realized by these optimal trajectories at time t are exactly ρ_t .

One can study the optimal control problem given by (8.23) and (8.24) by means of its value function.

Box 8.3. – Memo – Optimal control and Hamilton-Jacobi

Consider the optimization problem

$$\max \left\{ \int_0^T L(t, y(t), \alpha(t)) dt + \Phi(y(T)) : y' = f(t, y, \alpha), y(0) = x_0 \right\},$$

where α is the control that we can use to affect the solution of the state equation on the trajectory x . Define the so-called *value function*

$$\varphi(t, x) = \sup \left\{ \int_t^T L(s, y(s), \alpha(s)) ds + \Phi(y(T)) : y(t) = x, y' = f(s, y, \alpha) \right\}.$$

It is well-known from optimal control theory (see for instance Section 10.3 in [159]) that φ satisfies the Hamilton-Jacobi-Bellmann equation (in the viscosity sense)

$$\partial_t \varphi(t, x) + H(t, x, \nabla \varphi(t, x)) = 0, \quad \varphi(T, x) = \Phi(x),$$

where the Hamiltonian H is defined through

$$H(t, x, \xi) = \sup_{\alpha} \xi \cdot f(t, x, \alpha) + L(t, x, \alpha) \quad (8.25)$$

Moreover, it is also well-known that in the control problem, for every (t, x) , the optimal choice of $\alpha(t)$ so as to solve the optimization problem starting from x at time t is the control α which maximizes in the definition of H given in (8.25) for $\xi = \nabla \varphi(t, x)$, i.e. which maximizes $\nabla \varphi(t, x) \cdot f(t, x, \alpha) + L(t, x, \alpha)$.

In the remarkable case where $f(t, x, \alpha) = \alpha$ and $L(t, x, \alpha) = -\frac{1}{2}|\alpha|^2 - g(t, x)$ we get $H(t, x, \xi) = \frac{1}{2}|\xi|^2 - g(t, x)$, thus $\partial_t \varphi + \frac{1}{2}|\nabla \varphi|^2 = g$, and the optimal curves follow $y' = \nabla \varphi(t, y)$.

This gives a system of two coupled equations, since φ solves an HJB equation where ρ appears in the Hamiltonian, and ρ evolves according to a continuity equation $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$. To be more precise, we can give an explicit example, in the case $f(t, x, \alpha) = \alpha$. In such a case, the vector field to be put in the continuity equation is exactly $\nabla \varphi$. This gives the system

$$\begin{cases} \partial_t \varphi + \frac{|\nabla \varphi|^2}{2} - g(\rho) = 0, \\ \partial_t \rho + \nabla \cdot (\rho \nabla \varphi) = 0, \\ \varphi(T, x) = \Phi(x), \quad \rho(0, x) = \rho_0(x), \end{cases} \quad (8.26)$$

where the first equation is satisfied in the viscosity sense and the second in the distributional sense.

In this case, it is also known that a solution of this system (i.e. an equilibrium in the Mean Field Game) may be obtained by minimizing a suitable global functional (obviously, up to a change of sign, we could also express it as a maximization). Actually, one can solve

$$\min \int_0^T \int_{\Omega} \left(\frac{1}{2} |\alpha(t, x)|^2 \rho(t, x) + G(\rho(t, x)) \right) dx dt - \int_{\Omega} \Phi(x) \rho(T, x) dx$$

among solutions (ρ, α) of the continuity equation $\partial_t \rho + \nabla \cdot (\rho \alpha) = 0$ with initial datum $\rho(0, x) = \rho_0(x)$. When the function G is chosen as the antiderivative G of g (i.e. $G' = g$, and in particular G is convex), it happens that the minimizers of this global functional are equilibria in the sense explained above (the trajectories that the agents chose according to these densities are such that they exactly realize these densities). These functionals are a modification of the functional proposed by Benamou and Brenier [36] and studied in Chapters 5 and 6, and the minimization corresponds to finding a good curve in $\mathbb{W}_2$, which avoids over-congested measures. This kind of problems was also studied, without equilibrium issues, by Buttazzo, Jimenez and Oudet in [97]. Also note the analogy with the congested problems of Section 4.4.1. The main difference here is the crucial role of time, while the movement in Section 4.4.1 was stationary.

Semi-geostrophic equation One of the first applications of optimal transport has been its role in this equation modeling large scale atmospheric behavior in a certain asymptotical regime. The main ingredients of the physical description are the Coriolis force, the gravity, and an incompressibility constraint. We suppose that the particles could have different weights, thus leading to different impact of gravity, but their number per unit of volume should be constant in time and space. This consists in imposing that the density (mass per unit of volume) is constant in Lagrangian coordinates, but not necessarily in time and space.

By ignoring, in a very shallow atmosphere (consider that the radius of the Earth is much larger than the width of the atmosphere) the vertical effects of the Coriolis force, and using local coordinates where the vertical axis is orthogonal to the Earth surface, we can write an equation in Lagrangian formulation:

$$X''(t, a) + JX'(t, a) + \nabla p_t(X(t, a)) = f(a),$$

where p_t is the pressure at time t , a is a Lagrangian label, and the matrix $J \in \mathbb{M}^3 \times 3$ is given by

$$J = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The force f stands for gravity, is vertical, and only depends on the label a . The pressure p_t appears in order to take care of the incompressibility constraint, which corresponds to $a \mapsto X(t, a)$ being measure-preserving for every t . For details on this theory we refer the reader to the book by M. Cullen, [131].

In Eulerian coordinate the above equation is written

$$\begin{cases} \partial_t \mathbf{v}_t + (\mathbf{v}_t \cdot \nabla) \mathbf{v}_t + J \mathbf{v} + \nabla p_t = f_t, \\ \partial_t f_t + \mathbf{v}_t \cdot \nabla f_t = 0, \\ \nabla \cdot \mathbf{v}_t = 0, \end{cases}$$

where the second equation represents the fact that the force is transported by the flow itself (hence it “follows” the particles) and the third stands for incompressibility.

The time-scale regime of the model is quasi-stationary, in the sense that we suppose that the configuration at time t is stable. This can be interpreted in several ways. A first one, with a variational flavor, is the following: at each time t we have a distribution of particles, each with its velocity. Suppose that these particles want to move, each from a position x to x' . The new distribution of positions must be incompressible and, among these new redistributions, we need to impose that the original one minimizes the energy of the system, which is given by kinetic energy and potential energy. Consider that, because of Coriolis force, when moving from x to x' , the velocity moves from $\mathbf{v}(x)$ to $\mathbf{v}(x) + J(x' - x)$. Hence the energy to be minimized is of the form $\int \frac{1}{2} |J(x' - x)|^2 + \ell(x, x' - x)$, where $\ell(x, \cdot)$ is affine for every x (the gravity part is included in ℓ). Consider the pressure, which takes care of the incompressibility constraint. This means that minimizing such an energy under prescribed image measure (which is by the way an optimal transport problem with a cost equivalent to the quadratic one), is the same (see also the part on incompressible Euler) as minimizing $\int \frac{1}{2} |J(x' - x)|^2 + p(x') + \ell(x, x' - x)$ under no constraints. It is clear that each position is stable if and only if it minimizes $x' \mapsto \frac{1}{2} |J(x' - x)|^2 + p(x') + \ell(x, x' - x)$. This imposes first-order conditions, but also the fact that $x' \mapsto \frac{1}{2} |Jx'|^2 + p(x')$ is convex.

Another way of seeing the same fact is the following. Fix a time t_0 and a point x_0 , where particles have a velocity $\mathbf{v}_0$. Suppose that some particles around x_0 have a small perturbation in their speed $\mathbf{v} \approx \mathbf{v}_0$. We want to see what happens to these particles: do they stay close to x_0 or do they move far away? We freeze the pressure in time, and approximate it around x_0 with its Taylor expansion at x_0 . Consider a particle starting from x_0 with velocity $\mathbf{v}(s)$: here the variable s denotes time, but at a shorter time scale than that of the global evolution in t (which justifies the fact that we freeze $t = t_0$). The equation on $\mathbf{v}$, if we differentiate the acceleration law, is of the form $\mathbf{v}'' + J\mathbf{v}' + D^2 p(x_0)\mathbf{v} = 0$ (here we take a second-order Taylor expansion for p). By replacing $\mathbf{v}'$ with $-J\mathbf{v} + \text{const}$ (taking, for this lower-order condition, a first-order Taylor expansion for p) we get

$$\mathbf{v}'' + (D^2 p(x_0) - J^2)\mathbf{v} = \text{const}.$$

For stability, we want to avoid the existence of solutions of this equation which diverge exponentially. If only bounded solutions exist, this means that small perturbations in the speed produce small displacements, and hence the profile at (t_0, x_0) is stable.

In both interpretations (the variational one above, and the one related to linear stability), the condition that we find is that $D^2 p(x_0) - J^2$ (which is a symmetric matrix)

only has positive eigenvalues. This means

$$D^2p + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \geq 0$$

in the sense of positive-definite symmetric matrices. Equivalently, the function $x \mapsto P_t(x) := p_t(x) + \frac{1}{2}(|x_1|^2 + |x_2|^2)$ must be convex.

This convexity fact is known as the Cullen-Purser stability condition, see [131] or [286, 132].

We now come back to the acceleration law, we differentiate it as above, and replacing X'' using the equation we get

$$X'''(t, a) + [\nabla_t p(X(t, a))]' - J^2 X'(t, a) = J \nabla p_t(X(t, a)).$$

If we decide, on this time-scale, to ignore the third derivative (yep!), and we set $Y_t(a) = \nabla P_t(X(t, a))$, we have

$$Y'(t, a) = J(Y(t, a) - X(t, a)),$$

and this is an evolution equation on Y where X is defined from Y in the following way: for each time t , $X(t, \cdot)$ is measure-preserving, and Y is decomposed into the composition of the gradient of a convex function P and X . Here is where optimal transport, and in particular polar factorization (see Section 1.7.2) comes into play! We refer to [34] for the first applications of optimal transport in this field.

This equation is also studied in the so-called dual variables, i.e. Y and its image measure ρ (which has nothing to do with the density of the fluid). They solve

$$\begin{cases} \partial_t \rho_t + \nabla \cdot (\rho_t J(y - \nabla P_t^*(y))) = 0, \\ \det(D^2 P_t^*) = \rho_t, P_t^* \text{ is convex.} \end{cases}$$

This equation has been mathematically studied in several papers, where recent regularity improvements on the Hessian of the Kantorovich potential (the $W^{2,1}$ result that we briefly addressed in Section 1.7.6) have allowed for rigorous results (see for instance [14, 15]).

Reconstruction of the Early Universe This is another surprising domain where optimal transport for a quadratic cost played an important role. The problem considers the evolution of the density ρ of the matter in the Universe, subject to the gravity field created by ρ itself. In the so-called *co-moving coordinates* the equations read as follow

$$\begin{cases} \partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{3}{2} \frac{\nabla \phi + \mathbf{v}}{t}, \\ \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \\ \Delta \phi = \frac{\rho - 1}{t}. \end{cases} \quad (8.27)$$

The first equation gives the acceleration of a particle in terms of the gradient of the potential, the second is conservation of mass, and the third is the relation of the potential with the density itself. To understand where the precise expressions of the first and

third equations come from, the reader is invited to look at the very first pages of [47]. He will easily see that the main point is given by the change of variable imposed in the choice of the co-moving coordinates, i.e. coordinates which follow the expansion of the universe. In these coordinates we use $x' = a(t')x$ as a space variable, where x is the physical space variable and a is the expansion coefficient, which grows in time. We change the time scale from t to t' , via $dt = a(t')dt'$. After the change-of-variables, we drop the superscripts ' in the new variables. The expansion factor a needs to satisfy precise equations, called Friedmann equations. Note that also the friction term in the first equation (i.e. the negative velocity at the r.h.s.) exactly comes from this change-of-variable.

One sees from these equations that the density ρ at time 0 must be uniform. It corresponds to the distribution of matter in the very first instants of the universe. On the contrary, the density of the universe today is far from being uniform, and is well approximated by a number of Dirac masses corresponding each to a galaxy cluster (that is, each atom represents several galaxies, very close to each other). The question arises whether we can guess the evolution of the universe in the past by only knowing the current density ρ_T (T being the final time, i.e. now, but we will set $T = 1$ for simplicity), i.e. without knowing the initial velocity. This question has been analyzed by U. Frisch and collaborators, involving among others Y. Brenier because of the connections this problem has with optimal transport. See [90] and [173]. By the way, Appendix A in [90] explains in details the derivation of the above equations in co-moving coordinates presenting the set of choices and simplifications that have been made compared to the general framework in [47].

The connections of these equations and of the problem of the reconstruction from the only knowledge of ρ_T with optimal transport theory are twofold. First, let us start from the so-called Zel'dovich approximation. This approximation consists in assuming that, for $t > 0$, all the particles move on straight lines with constant speed only depending on the initial data. Let us be more precise: in Lagrangian coordinates the first equation of (8.27) reads

$$\frac{2t}{3}X''(t, x_0) + X'(t, x_0) = -\nabla\varphi_t(X(t, x_0)).$$

Here the potential φ_t is required to satisfy $1 + t\Delta\varphi_t = \rho_t$, where ρ_t is the image measure of $X(t, \cdot)$. The approximation consists in ignoring the second derivative, and freezing the r.h.s. at its value at $t = 0$, i.e. $X(t) = x_0 - t\nabla\varphi_0(x_0)$.

It seems a rude approximation, but it gives good results in practice, and it has an optimal transport justification, as pointed out by Y. Brenier. Indeed, choose as ρ_t the geodesic in $\mathbb{W}_2$ between $\rho_0 = 1$ and ρ_1 . If $-\varphi_t$ is the Kantorovich potential between ρ_t and ρ_0 , we know that it solves the Monge-Ampère equation

$$\det(I + tD^2\varphi_t) = \rho_t.$$

Moreover, if T is the optimal transport from ρ_0 to ρ_1 , the Lagrangian map $X(t, x_0) = (1-t)x_0 + tT(x_0)$ is the optimal transport from ρ_0 to ρ_t . This means that particles should arrive at time t with velocity equal to $-\nabla\varphi_t(X(t, x_0))$ and proves that $-\nabla\varphi_t(X(t, x_0)) = T(x_0) - x_0$ does not depend on time. In particular, it is also equal to its initial value

$-\nabla\varphi_0(x_0)$. In this way, we have constructed a Lagrangian map compatible with the Zel'dovich approximation (i.e. particles move with $X'' = 0$), with the only exception that it solves $\det(I + tD^2\varphi_t) = \rho_t$ instead of $1 + t\Delta\varphi_t = \rho_t$. In this way, the Zel'dovich approximation provides approximate solution for the original equation but exact solutions for this new, modified equation. Note the use of a nonlinear equation to approximate a linear one (the linear one being the linearization of the non-linear around $t = 0$), quite peculiar in this case.

From the above observations, the authors of [90] concentrated on the search for an optimal map T in the transport between the uniform density ρ_0 and the current density ρ_1 and obtained satisfactory reconstruction results. Numerically, they discretized also the source measure and used an auction algorithm.

On the other hand, [90] also shows (in Section 6 and Appendix D) that System (8.27) (without approximation or replacing the Poisson equation with the Monge-Ampère equation) can be obtained as the optimality condition of a variational problem, namely

$$\min \left\{ \int_0^1 \int \left(\frac{t^{3/2}}{2} \rho |\mathbf{v}|^2 + \frac{3t^{-1/2}}{4} |\nabla\varphi|^2 \right) \text{ with } \begin{cases} \partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \\ \Delta\varphi = \rho - 1, \\ \rho_0 = 1, \rho_1 = \text{given} \end{cases} \right\}. \quad (8.28)$$

This problem is a sort of modified Benamou-Brenier problem, with a time-scale factor, and the addendum of a term $F(\rho)$ which penalizes (with a time-scale coefficient as well) the H^{-1} norm of $\rho - 1$. Note that this term is the same which appears in chemotaxis gradient flows (see Section 8.4.2), but with the good sign, and also note the same formalism as the variational formulation of Mean Field Games. Moreover, the important fact is that this minimization problem is strictly convex, which implies uniqueness of the solution of (8.27).

We leave the proof of the fact that the optimality conditions of the above minimization lead to (8.27) as an exercise, **Ex(54)**.

We also stress that the existence of a minimizer is not straightforward. Indeed, the coefficient $t^{3/2}$ in front of the kinetic term does not allow to preserve the condition at $t = 0$ along minimizing sequence, and the H^{-1} term is multiplied times a coefficient $t^{-1/2}$ which is integrable, hence not enough to compensate for it. We are in a situation analogous to the following 1D problem

$$\min \left\{ \int_0^1 t^{1+\alpha} |x'(t)|^2 + t^{-1+\alpha} G(x(t)) : x(1) = x_1 \right\}$$

where G is a convex functional, with quadratic growth, minimal at x_0 . We omit the constraint $x(0) = 0$ since this condition would not be stable under convergence. In particular, there are finite-energy curves $x(t)$ with $x(0) \neq x_0$ (compare to **Ex(56)**). On the other hand, one can prove that the minimizers of this problem must satisfy an extra condition at $t = 0$, because of optimality (like Neumann conditions, which are not meaningful here since they only give $0 = 0$ because of the coefficient $t^{3/2}$). It is possible to see, as in **Ex(55)**, that this condition is $x(0) = x_0$.

The minimal action problem (8.28) has also been studied, in a simplified setting, by Loeper in [216]. We also mention a variant problem proposed by Brenier, which

is meant to could give a better insight in the formation of concentration, [88]. Indeed, it is quite disappointing to think that in the optimal transport problem between ρ_0 and ρ_1 , Dirac masses only appear at the final time (coherently with the analysis of the intermediate measures μ_t that we did in Section 4.3). This means that they only appear now... and if we restudied again the same problem tomorrow?

Finally, it is challenging to think that both the “static” version of this cosmological problem (i.e. the optimal transport between ρ_0 and ρ_T) and the “dynamical” one (8.28) still have many secrets to reveal, if treated with the new numerical methods and approaches (see Chapter 6, in particular for semi-discrete methods which are exactly suitable for $\rho_0 = 1$ and atomic ρ_T) which were not available at the time of [90].

Exercises

The following is a short list of possible exercises related to the subject of the book. The level of difficulty is strongly heterogeneous and the most difficult exercises are indicated with one * or two **. The symbol ♠ is associated to those exercises which are more relevant for the theory. Some (very) short hints follow.

Exercises from the topics of Chapter 1

Exercise 1. Find an example of a sequence of functions $f_n : [0, 1] \rightarrow [0, 1]$ such that $(f_n)_\#(\mathcal{L}^1 \llcorner [0, 1]) = \mathcal{L}^1 \llcorner [0, 1]$ but $f_n \rightarrow \frac{1}{2}$. Can these functions be taken C^1 ?

Exercise 2. ♠ Consider the problem

$$\max \left\{ \int |x - T(x)|^2 d\mu \quad : \quad T_\# \mu = \nu \right\}.$$

Prove that, if μ is absolutely continuous, it admits a solution and the optimal T is the gradient of a concave function.

Exercise 3. Find the optimal transport map for the quadratic cost $c(x, y) = |x - y|^2$ between $\mu = f \cdot \mathcal{L}^2$ and $\nu = g \cdot \mathcal{L}^2$ in 2D, where $f(x) = \frac{1}{\pi} \mathbb{1}_{B(0,1)}(x)$ and $g(x) = \frac{1}{8\pi}(4 - |x|^2)$.

Exercise 4. Let $R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be given by $R(x) = -x$. Characterize the probabilities $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ such that R is an optimal transport map between μ and $R_\# \mu$ for the quadratic cost.

Exercise 5. Let $S : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be given by $S(x_1, x_2, \dots, x_d) = (x_1, x_2, \dots, -x_d)$. Prove that S cannot be the optimal transport map between μ and $S_\# \mu$ for the quadratic cost if $\mu \ll \mathcal{L}^d$. Prove that, if S is optimal in the above sense, then μ is concentrated on a Lipschitz hyper-surface of the form $\{x_d = f(x_1, x_2, \dots, x_{d-1})\}$, with $f \in \text{Lip}_1(\mathbb{R}^{d-1})$.

Exercise 6. Consider the multi-marginal problem

$$\min \left\{ \int c(x_1, \dots, x_N) d\gamma : \gamma \in \mathcal{P}((\mathbb{R}^d)^N) (\pi_i)_\# \gamma = \mu_i \right\},$$

where $\mu_i \in \mathcal{P}(\mathbb{R}^d)$ are given compactly supported measures. Suppose that γ is concentrated on a set $\{(x_1, \dots, x_N) \in (\mathbb{R}^d)^N : x_1 + \dots + x_N = \text{constant}\}$. Prove that γ is optimal whenever the cost c is given by $c(x_1, \dots, x_N) = h(x_1 + \dots + x_N)$ for a convex function $h : \mathbb{R}^d \rightarrow \mathbb{R}$ and also in the case $c(x_1, \dots, x_N) = -\sum_{i \neq j} |x_i - x_j|^2$.

Exercises from the topics of Chapter 2

Exercise 7. Consider two probability measures μ and ν on $\mathbb{R}$, that we will suppose atomless and with full support for simplicity. Let F_μ and F_ν be their cumulative distribution functions: $F_\mu(x) = \mu((-\infty, x])$ and $F_\nu(x) = \nu((-\infty, x])$. Prove that the following facts are equivalents

1. $F_\mu \geq F_\nu$,
2. there exists $\gamma \in \Pi(\mu, \nu)$ concentrated on $\{(x, y) : y \geq x\}$,
3. there exists $T : \mathbb{R} \rightarrow \mathbb{R}$ such that $T_\# \mu = \nu$ and $T(x) \geq x$ for every x ,
4. the monotone increasing transport T from μ to ν satisfies $T(x) \geq x$ for every x .

Exercise 8. Prove that an optimal $\gamma \in \pi(\mu, \mu)$ for the cost $c(x, y) = \frac{1}{|x-y|}$ exists, and that it is concentrated on at most two increasing graphs, one above and one below the diagonal. Prove that these two graphs do not superpose and that the plan comes indeed from a transport map. Is it increasing?

Exercise 9. * In the case $\mu = \mathcal{L}^1 \llcorner [0, 1]$ find the optimal transport map from μ to μ for the cost $c(x, y) = \frac{1}{|x-y|}$.

Exercise 10. ♠ Prove the optimality of γ_{mon} in 1D under the assumption that the cost $c : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is C^2 with $\frac{\partial^2}{\partial x \partial y} c(x, y) < 0$ for every $(x, y) \in \mathbb{R} \times \mathbb{R}$. Deduce the existence of an optimal map if μ is atomless.

Exercise 11. Let A and B two symmetric positive definite matrices and $L_A, L_B : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be affine functions defined through $L_A(x) = Ax + a$, $L_B(x) = Bx + b$, for given $a, b \in \mathbb{R}^d$. Take a reference measure ρ which is invariant under rotations (i.e. $R_\# \rho = \rho$ for every matrix R s.t. $R \cdot R^t = I$) and define $\mu := (L_A)_\# \rho$ and $\nu := (L_B)_\# \rho$. Find the optimal transport map from μ to ν for the quadratic cost.

Exercises from the topics of Chapter 3

Exercise 12. Consider a domain $\Omega = [0, 3] \times [0, 2]$ and two absolutely continuous measures on it, μ_0 and μ_1 , with density ρ_0 and ρ_1 , respectively, given by

$$\begin{aligned} \rho_0(x, y) &= \frac{2}{9} \text{ if } x \leq 1 \text{ or } x \in]1, 2[\text{ and } y \leq 1, \\ \rho_0(x, y) &= \frac{1}{9} \text{ if } x \geq 2 \text{ or } x \in]1, 2[\text{ and } y > 1, \\ \rho_1(x, y) &= \frac{2}{9} \text{ if } x \geq 2 \text{ or } x \in]1, 2[\text{ and } y \leq 1, \\ \rho_1(x, y) &= \frac{1}{9} \text{ if } x \leq 1 \text{ or } x \in]1, 2[\text{ and } y \geq 1. \end{aligned}$$

Represent graphically these two densities and find an optimal transport, as well as a Kantorovich potential, between μ_0 and μ_1 for the cost $c(x, y) = |x - y|$. Is the optimal transport unique? and the potential?

Exercise 13. Find the optimal transport map for the cost $c(x, y) = |x - y|$ between $\mu = f \cdot \mathcal{L}^2$ and $\nu = g \cdot \mathcal{L}^2$ in 2D, where $f(x) = \frac{1}{\pi} \mathbb{1}_{B(0,1)}(x)$ and $g(x) = \frac{1}{8\pi}(4 - |x|^2)$.

Exercise 14. Consider the cost $c(x, y) = |x - y|^\alpha$ for a given exponent $0 < \alpha < 1$. Prove that, for every pair $\mu, \nu \in \mathcal{P}(\Omega)$, we have

$$\mathcal{T}_c(\mu, \nu) = \sup \left\{ \int u d(\mu - \nu) : u \in C^{0,\alpha}(\Omega), [u]_{0,\alpha} \leq 1 \right\},$$

where $[u]_{0,\alpha}$ is the $C^{0,\alpha}$ seminorm of u : $[u]_{0,\alpha} := \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha}$.

Exercise 15. Consider the cost $c(x, y) = \arctan(|x - y|)$ and two measures $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ with compact and disjoint supports.

1. What can we say about c -concave functions for this c ? are they (uniformly) continuous, Lipschitz...?
2. Do the problems $\min \left\{ \int c(x, y) d\gamma : \gamma \in \Pi(\mu, \nu) \right\}$ admit a solution?
3. Write the dual problem in terms of functions u having a given modulus of continuity.
4. Which is the connection between the support of an optimal γ and the optimal u in the dual problem?
5. Supposing that μ is absolutely continuous, prove that the optimal γ is unique and derives from a transport. Give its expression in terms of ∇u .

Exercise 16. Consider $h : \mathbb{R}^d \rightarrow \mathbb{R}$ defined through $h(z) = (|z| - 1)_+^2$. Prove that, for every $\mu, \nu \in \mathcal{P}(\Omega)$ with $\mu \ll \mathcal{L}^d$ and Ω compact, there exists an optimal transport map for the cost $c(x, y) = h(x - y)$.

Exercise 17. Consider the cost $c(x, y) = \lfloor |x - y| \rfloor$ (the integer part of the distance: $\lfloor a \rfloor = \max\{n \in \mathbb{Z} : n \leq a\}$). Prove that, for every $\mu, \nu \in \mathcal{P}(\Omega)$ with $\mu \ll \mathcal{L}^d$ and Ω compact, there exists an optimal transport plan for such a cost of the form $\gamma = \sum_{i=1}^N \gamma_i$ where each γ_i is induced by a transport map and $N \leq \text{diam}(\Omega) + 1$.

Exercise 18. For $\varepsilon > 0$, consider the cost $c_\varepsilon(x, y) = \varepsilon \left\lfloor \frac{|x - y|}{\varepsilon} \right\rfloor$ and let γ_ε be an optimal transport plan for such a cost between two measures $\mu, \nu \in \mathcal{P}(\Omega)$ (consider Ω compact for simplicity). Prove that, up to subsequences, $\gamma_\varepsilon \rightarrow \gamma$, where γ is an optimal transport plan between the same measures for the cost $c(x, y) = |x - y|$.

Exercise 19. ♠ Consider the cost $c_\varepsilon(x, y) = \sqrt{\varepsilon^2 + |x - y|^2}$ and let γ_ε be optimal for the cost c_ε in $\Pi(\mu, \nu)$. Let $\varepsilon \rightarrow 0$. Prove that, up to subsequences, we have $\gamma_\varepsilon \rightarrow \gamma$ where γ is optimal for the cost $|x - y|$. Supposing $\mu \ll \mathcal{L}^d$ and $\text{spt}(\mu) \cap \text{spt}(\nu) = \emptyset$, prove that we have full convergence to the ray-monotone transport plan. What can we say about the limit $\varepsilon \rightarrow \infty$?

Exercise 20. * Find a Borel set $A \subset [0, 1]$ such that for every interval $I \subset [0, 1]$ we have $0 < \mathcal{L}^1(I \cap A) < \mathcal{L}^1(I)$.

Exercise 21. Let us define a distance, called *SNCF distance*, on a set X in the following way:

$$\text{SNCF}(x, y) := \begin{cases} 0 & \text{if } x = y, \\ P(x) + P(y) & \text{if } x \neq y, \end{cases}$$

where $P : X \rightarrow \mathbb{R}$ is a bounded function, to be interpreted as the distance to a fixed connecting hub: imagine that every point is connected by a direct train to this hub, and to no other location⁴. Prove that $\text{SNCF}(x, y)$ is a distance on X and that, for $c(x, y) = \text{SNCF}(x, y)$ and $\mu, \nu \in \mathcal{P}(X)$, we have $\mathcal{T}_c(\mu, \nu) = \int P(x) d|\mu - \nu|(x)$.

Exercises from the topics of Chapter 4

Exercise 22. Let y be a solution of the ODE $y'(t) = \mathbf{v}_t(y(t))$. Prove that $\rho_t := \delta_{y(t)}$ is a solution of the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}_t) = 0$.

Exercise 23. * Prove that the set of polynomial functions is dense for the C^1 norm in the set $C^1([0, 1]^d)$.

Exercise 24. Find the transport density σ (see Section 4.3) between the measures $\mu = f \cdot \mathcal{L}^2$ and $\nu = g \cdot \mathcal{L}^2$ in 2D, where $f = \frac{1}{\pi} \mathbb{1}_{B(0,1)}$ and $g(x) = \frac{1}{4\pi} \mathbb{1}_{B(0,2)}$.

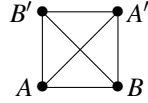
Exercise 25. Suppose $\mu \in L^p(\Omega)$ and $\nu \in L^q(\Omega)$, with $p > q$. Prove that, if $p < d/(d-1)$, the transport density σ in the transport from μ to ν belongs to L^p and, if $p \geq d/(d-1)$, it belongs to $L^r(\Omega)$ for all the exponents r satisfying

$$r < r(p, q, d) := \frac{dq(p-1)}{d(p-1) - (p-q)}.$$

Exercise 26. ♠ Given $\mu, \nu \in \mathcal{P}(\Omega)$, prove that there exists $Q \in \mathcal{P}(\mathcal{C})$ with $(e_0)_\# Q = \mu$, $(e_1)_\# Q = \nu$ and $i_Q \in L^p(\Omega)$ (we recall that i_Q is the traffic intensity defined in Section 4.2.3 and that the e_t are the evaluation maps) if and only if $\mu - \nu \in (W^{1,p'})^*$.

Exercise 27. Given $Q \in \mathcal{P}(\mathcal{C})$ with $(e_0)_\# Q = \mu$, $(e_1)_\# Q = \nu$, prove that there exists $\tilde{Q} \in \mathcal{P}(\mathcal{C})$ concentrated on injective curves and such that $(e_0)_\# \tilde{Q} = \mu$, $(e_1)_\# \tilde{Q} = \nu$ and $i_{\tilde{Q}} \leq i_Q$.

Exercise 28. Consider the (simple) network on the right, where the cost for a commuter to move along a segment of length ℓ with traffic intensity i is given by $i\ell$. Do not distinguish between different traffic directions on a same segment (i.e. i is the sum of the number of commuters going in one direction plus the number going in the other direction). Consider the case where a quantity 1 of commuters must go from A to A' and another quantity 1 from B to B' . Find the Wardrop equilibrium.



Exercise 29. Prove that $d_\alpha(\mathcal{L}_\Omega, \delta_0) = +\infty$ for every compact domain $\Omega \subset \mathbb{R}^d$ with non-empty interior and $|\Omega| = 1$, and every $\alpha < 1 - 1/d$.

⁴The common hub could be called Paris, for instance. However, no restriction on the space X is imposed, and in particular there is no need to assume that it has hexagonal shape.

Exercise 30. Consider the branched transport for $\alpha = 1/2$ between two atomic measures $\mu = \sum_i a_i \delta_{x_i}$ and $\nu = \sum_j b_j \delta_{y_j}$. Consider a triple junction at a point z which is not one of the points x_i or y_j : among the three branches at z , either two are incoming and one out coming or two out coming and one incoming. Prove that the two with the same orientation must make an angle of 90° .

Exercises from the topics of Chapter 5

Exercise 31. ♠ Let X be a Polish metric space, and $\mu_n, \mu \in \mathcal{P}_1(X)$ be probability measures on X such that $\int \phi d\mu_n \rightarrow \int \phi d\mu$ for every $\phi \in \text{Lip}_1(X)$. Prove $\int \phi d\mu_n \rightarrow \int \phi d\mu$ for every $\phi \in C_b(X)$.

Exercise 32. Prove the following equalities, for every compact set $\Sigma \subset \Omega$ and every density $f \in L^1(\Omega)$ with $f \geq 0$ and $\int_\Omega f(x) dx = 1$,

$$\begin{aligned} \int d(x, \Sigma) f(x) dx &= \min\{W_1(f, \nu) : \nu \in \mathcal{P}(\Sigma)\} \\ &= \min\{|\mathbf{w}|(\Omega) : \mathbf{w} \in \mathcal{M}_{\text{div}}^d(\Omega), \text{spt}(\nabla \cdot \mathbf{w} - f) \subset \Sigma\}. \end{aligned}$$

What could we say if we considered, instead, $\int d(x, \Sigma)^p f(x) dx$, for $p > 1$?

Exercise 33. Let $\gamma \in \Pi(\mu, \nu)$ be a transport plan between two probabilities on two compact spaces X and Y , respectively. Let $\mu_n \rightharpoonup \mu$ and $\nu_n \rightharpoonup \nu$ be two weakly converging sequences of probabilities. Prove that there exists a sequence $\gamma_n \in \Pi(\mu_n, \nu_n)$ with $\gamma_n \rightharpoonup \gamma$. Deduce that for any sequence of costs c_n converging uniformly to c we have $\mathcal{T}_{c_n}(\mu_n, \nu_n) \rightharpoonup \mathcal{T}_c(\mu, \nu)$.

Exercise 34. Let $\Omega \subset \mathbb{R}^d$ be smooth and compact, and V be an operator which associates to every probability $\rho \in \mathcal{P}(\Omega)$ a vector field $V[\rho] : \Omega \rightarrow \mathbb{R}^d$ with the following properties: for every ρ , $V[\rho]$ is Lipschitz continuous and satisfies $\text{Lip}(V[\rho]) \leq C$ (for a constant C not depending on ρ) and $V[\rho] \cdot \mathbf{n} = 0$ on $\partial\Omega$; the map $\rho \mapsto V[\rho]$ is Lipschitz continuous in the following sense: $\|V[\rho] - V[\mu]\|_{L^\infty} \leq CW_2(\rho, \mu)$ for every ρ, μ . Consider the equation $\partial_t \rho + \nabla \cdot (\rho V[\rho]) = 0$ with initial Cauchy datum $\rho_0 \ll \mathcal{L}^d$. Prove that any solution ρ_t stays absolutely continuous for $t > 0$ and prove uniqueness of the solution.

Exercise 35. ♠ Prove the equality $W_\infty(\mu, \nu) = \lim_{p \rightarrow \infty} W_p(\mu, \nu)$ for μ, ν compactly supported. Prove also that for any sequence $p_n \rightarrow \infty$, the optimal plans γ_n for the cost $|x - y|^{p_n}$ converge, up to subsequences, to an optimal plan for the problem defining the W_∞ distance.

Exercise 36. Prove the inequality $d_H(\text{spt}(\mu), \text{spt}(\nu)) \leq W_\infty(\mu, \nu)$, where d_H denotes the Hausdorff distance.

Exercise 37. * Is it true that for $1 \leq p \leq q \leq \infty$, if the same map T is optimal in the definition of W_p and W_q , then it is also optimal in the definition of W_r for every $r \in [p, q]$?

Exercise 38. Prove that if $f \in W^{1,p}(\Omega)$, Ω is a convex set, $\mu, \nu \in \mathcal{P}(\Omega) \cap L^r(\Omega)$ and $\|\mu\|_{L^r}, \|\nu\|_{L^r} \leq C$, then, for $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1 + \frac{1}{qr}$, one has

$$\int_{\Omega} f d(\mu - \nu) \leq C^{\frac{1}{q}} \|\nabla f\|_{L^p(\Omega)} W_q(\mu, \nu).$$

Exercises from the topics of Chapter 6

Exercise 39. Let $\Sigma = \{x_1, \dots, x_N\} \subset \Omega$ be a finite subset of a compact convex domain Ω . For every i , let us set $V_i := \{x \in \Omega : |x - x_i| \leq |x - x_j| \text{ for all } j\}$. Suppose that Σ solves the problem $\min\{\int_{\Omega} d(x, \Sigma)^2 dx : \#\Sigma \leq N\}$. Prove that, for every i , the point x_i is the barycenter of V_i .

Exercise 40. ♠ For $A \subset \mathbb{R}^d$ we define the extremal points of A as those points $x \in A$ such that $y, z \in A, x = \frac{y+z}{2} \Rightarrow y = z$. Let $K \subset \mathcal{M}_n(\mathbb{R})$ be the set of bistochastic matrices (those matrices $M = (m_{ij})$ with $m_{ij} \geq 0$, $\sum_j m_{ij} = \sum_i m_{ij} = 1$ for all i, j). Prove that the extremal points of this set are the permutation matrices.

Exercise 41. For $a \in (\mathbb{R}_+)^n$ and $b \in (\mathbb{R}_+)^m$ consider the set $\Pi(a, b)$ defined through $\Pi = \{M \in \mathcal{M}_{n,m}(\mathbb{R}) : m_{ij} \geq 0, \sum_j m_{ij} = a_i, \sum_i m_{ij} = b_j\}$. Prove that every extremal matrix M in $\Pi(a, b)$ is such that $\#\{(i, j) : m_{ij} > 0\} \leq n + m$.

Exercise 42. Let $\gamma_{ij}^\varepsilon := p_i^\varepsilon q_j^\varepsilon \eta_{ij}$ be a solution of the approximated problem in (6.3) (see Section 6.4.1). Suppose that $\varepsilon \log(p_i^\varepsilon)$, $\varepsilon \log(q_j^\varepsilon)$ and γ_{ij}^ε converge as $\varepsilon \rightarrow 0$. Prove that the limit of $\varepsilon \log(p_i^\varepsilon)$ and $\varepsilon \log(q_j^\varepsilon)$ are Kantorovich potentials in the limit problem for $\varepsilon = 0$.

Exercise 43. Let u be a smooth function on Ω and $M(x) := \text{cof}(D^2 u(x))$. Let M^{ij} be the (i, j) -entry of M and M_k^{ij} its derivative w.r.t. x_k . Prove that for every index j we have $\sum_i M_i^{ij}(x) = 0$.

Exercise 44. * ♠ Consider a smooth transport map $T : \Omega \rightarrow \mathbb{R}^d$ of the form $T = \nabla u$ on a smooth domain Ω . Suppose $T_{\#} f = \nu$, where f is a smooth positive density on Ω . For every vector field $\mathbf{v} : \Omega \rightarrow \mathbb{R}^d$ with $\nabla \cdot (f \mathbf{v}) = 0$ and $\mathbf{v} \cdot \mathbf{n} = 0$ on $\partial\Omega$ consider the flow of the autonomous vector field $\mathbf{v}$, given by $Y_t(x) = y_x(t)$ (where $y'_x(t) = \mathbf{v}(y_x(t))$ and $y_x(0) = x$) and define $T_t = T \circ Y_t$. Let $M(S) = \int |S(x) - x|^2 f(x) dx$. Prove that the function $j(t) = M(T_t)$ satisfies $j''(0) \geq 0$ if and only if u is convex.

Exercises from the topics of Chapter 7

Exercise 45. * ♠ Prove that the entropy functional

$$\mathcal{F}(\rho) := \begin{cases} \int_{\mathbb{R}^d} \rho(x) \log \rho(x) dx & \text{if } \rho \ll \mathcal{L}^d, \\ +\infty & \text{otherwise.} \end{cases}$$

is l.s.c. in $\mathbb{W}_1(\mathbb{R}^d)$. Also prove that if $\rho_n \in \mathcal{P}_2(\mathbb{R}^d)$ and $\rho_n \rightharpoonup \rho$, $\int |x|^2 d\rho_n(x) \leq C$, then $\mathcal{F}(\rho) \leq \liminf_n \mathcal{F}(\rho_n)$.